

Lottery matching game from CS 170 Lab 4

For this assignment, the use of generative AI tools (including ChatGPT, GitHub Copilot, Claude, Gemini, and similar code-generation tools) is **not permitted** at any stage — including generating, completing, debugging, or explaining your code.

Write a program `Lottery.java` that randomly generates a lottery of a two-digit (10-99) number, prompts the user to enter a two-digit number, and determines whether the user wins according to the following rule:

If the user input matches the lottery in exact order, the award is \$10,000.

If the user input matches the lottery, the award is \$3,000.

If one digit in the user input matches a digit in the lottery, the award is \$1,000.

Examples

Input / Scenario	Expected result	Why
lottery=33, user=33	Win \$10,000	exact match (33 == 33)
lottery=33, user=34	Win \$1,000	3 appears in lottery; no exact/partial
lottery=33, user=45	No prize	4 and 5 do not appear in lottery 33
lottery=34, user=34	Win \$10,000	exact match
lottery=34, user=43	Win \$3,000	digits 4,3 swapped to 3,4 — partial match
lottery=34, user=35	Win \$1,000	digit 3 appears in lottery 34
lottery=34, user=45	Win \$1,000	digit 4 appears in lottery 34
lottery=34, user=25	No prize	2 and 5 do not appear in lottery 34

Tasks (complete in order):

(10 pts) Generate the lottery number using random method in the Math class. Extract its tens and units digits using / and %. Add a comment showing your digit expressions.

(10 pts) Declare three boolean variables, `exactMatch`, `partialMatch`, `digitMatch`, and assign each the correct boolean expression using the digit variables.

(20 pts) Write the if-else if-else chain: check `exactMatch` first, then `partialMatch`, then `digitMatch`, and finally the else clause for no match. As you write it, consider: what happens if you do not follow this order strictly, for example if you check `digitMatch` before `exactMatch`? Trace through a specific case where the user's input exactly matches the lottery number, and follow your code line by line. Which prize would your program award, and is it correct?

(10 pts) Add a branch trace comment showing which boolean is true for the case `lottery=34, user=43`. Walk through each if condition step by step to explain which branch fires and why.